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Presence and agency in lifestyle medicine: A neuropsychobiological framework for sustainable health transformation

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ABSTRACT

In an era dominated by chronic disease, Lifestyle Medicine (LM) offers a profound paradigm shift, yet its efficacy is often tethered to sustained behavioral change. This article proposes a neuropsychobiological framework for LM centered on three interconnected pillars: Presence, Agency, and Connectivity. We argue that cultivating Presence—an attuned awareness of internal bodily states (interoception) and external environments—provides the foundational data for well-being. Agency, the belief in one's capacity to act intentionally and effectively upon this data, translates insight into behavior. Finally, Connectivity, encompassing both neural integration and social belonging, fosters resilience and sustains health transformations. Drawing on interoceptive predictive coding models, active inference, and the neuroscience of social bonding, we illuminate the underlying mechanisms by which these pillars empower individuals to transcend passive health management. Through humanized clinical examples and practical strategies, we illustrate how this framework can transform LM from prescriptive advice to an empowering journey of self-discovery and sustainable health.

1. Introduction

Reclaiming the Internal Compass for Health Imagine trying to navigate life without knowing what your body is truly telling you. That sudden flutter in your chest when stressed, the subtle cues that signal true hunger versus emotional eating, the quiet sensations that guide your intuition—these are all products of an intricate internal compass. This compass, our interoceptive system, provides the fundamental data for well-being. When this capacity falters, individuals can lose their internal guidance system, becoming disconnected from their physiological needs and emotional landscapes. And when they lose that compass, they lose something far more valuable: the ability to actively shape their own health journey (see Tables 1–3).

Lifestyle Medicine (LM) stands at the forefront of chronic disease prevention and reversal, advocating for evidence-based interventions in diet, physical activity, sleep, stress management, and social connection [1]. However, the chasm between knowing "what to do" and consistently "doing it" remains a significant challenge. This article posits that true, sustainable health transformation in LM transcends mere knowledge acquisition; it hinges on an individual's capacity for Presence, Agency, and Connectivity. These three neuropsychobiological pillars, when intentionally cultivated, form a powerful synergistic loop that empowers individuals to reclaim their internal compass and become active architects of their health. We will delve into the scientific underpinnings of each pillar, illustrate their clinical relevance through humanized examples, and propose practical strategies for their integration into LM practice.

2. The foundation: presence and the wisdom of interoception

Presence, in our framework, is the mindful, non-judgmental awareness of one's internal and external environments in the present moment. At its core, it is profoundly rooted in interoception: the dynamic process by which the nervous system senses, interprets, and integrates signals originating from within the body [2]. This isn't just about feeling your heartbeat; it's about the subtle shifts in your gut, the nuanced tension in your muscles, the changes in your breath, and how these internal sensations inform your emotional state and cognitive processes.

When we pause to truly listen to our body—that slight quiver in the hands before a big presentation, the heavy warmth in the stomach after a satisfying meal—we are engaging our interoceptive pathways. This process is not passive reception but an active, predictive inference

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mechanism [3]. Our brains are constantly generating models of what our internal state should be, based on past experiences and current context. When actual sensory input deviates from these predictions, an "interoceptive prediction error" is generated, prompting updates to our internal models and guiding adaptive responses [4]. For instance, the brain might predict mild hunger, but the stomach sends signals of intense emptiness; this prediction error drives the conscious urge to seek food.

Neurobiologically, the anterior insular cortex (AIC) stands as a critical hub for interoceptive processing, integrating visceral, somatic, and affective information [5]. It acts as a nexus where raw bodily sensations are transformed into subjective feelings and conscious awareness. The anterior cingulate cortex (ACC) also plays a vital role in monitoring conflict and regulating attention towards these internal states. Dysfunction in these circuits can lead to a fundamental disconnect—a person might feel anxious but be unable to pinpoint the physiological origins or even recognize the physical sensations of anxiety.

To truly understand interoception's role in LM, we must differentiate its three key dimensions.

1. Interoceptive Accuracy: The objective ability to detect and count internal bodily sensations, such as heartbeat (e.g., heart-beat counting task). This is the physiological "ground truth."
2. Interoceptive Sensibility: The self-reported awareness and confidence in one's ability to perceive internal sensations. This is the subjective "feeling" of one's body.
3. Interoceptive Awareness: A higher-order cognitive process involving metacognitive insight into one's interoceptive experiences, including their meaning and relevance to emotions and behavior [6]. This is the "understanding" of what the body is saying.

Cultivating Presence through enhanced interoception means improving all three dimensions. It's about not just feeling your body (sensibility) or accurately detecting its signals (accuracy), but understanding what those signals mean for your overall well-being and how they connect to your thoughts and emotions (awareness). This foundational self-knowledge is the wellspring from which intentional health behaviors can flow.

3. The driving force: agency and the power of intentional action

If Presence is about receiving the body's wisdom, Agency is about having the courage and capacity to act upon it. Agency reflects an individual's perceived control over their own actions and outcomes—an internal locus of control that empowers them to initiate and sustain health-promoting behaviors. It is the belief, deeply rooted in our neurological architecture, that "I can make a difference in my own health."

From a neuropsychological perspective, agency is closely linked to active inference, a concept where the brain doesn't just predict sensory input, but actively acts upon the world to minimize prediction errors [7]. If our internal models predict a certain outcome (e.g., "I will feel less stressed if I exercise"), then agency drives the active engagement with that behavior. This isn't just a passive "hope" for change; it's a proactive engagement with the environment to bring our internal states closer to our desired, healthy predictions.

The prefrontal cortex, particularly the dorsolateral prefrontal cortex (dlPFC) for planning and working memory, and the ventromedial prefrontal cortex (vmPFC) for value-based decision-making, are crucial for fostering agency [8]. These regions enable us to set goals, weigh consequences, and override impulsive urges in favor of long-term health objectives. When agency is strong, an individual can recognize their hunger cues (Presence), and then intentionally choose a nourishing meal over a highly processed one, resisting immediate gratification.

Furthermore, the concept of self-efficacy, as defined by Bandura [9], is a cornerstone of agency. It's the belief in one's own ability to succeed in specific situations or accomplish a task. High self-efficacy in LM translates to confidence in one's ability to maintain a healthy diet, engage in regular physical activity, or manage stress effectively. This conviction is not merely a psychological construct; it's wired into our reward pathways and dopaminergic systems, which reinforce successful actions and motivate future efforts [10]. When someone successfully navigates a craving or achieves a fitness goal, their brain releases dopamine, strengthening the neural pathways associated with that positive behavior and increasing their sense of agency for future challenges [11,12].

Conversely, a diminished sense of agency, often observed in chronic illness or prolonged stress, can lead to learned helplessness [13]. Individuals might feel that their efforts are futile, leading to apathy and disengagement from health-promoting behaviors. This sense of powerlessness can be highly debilitating, trapping individuals in cycles of poor health despite their knowledge of healthier choices. Therefore, fostering agency in LM is not about demanding compliance; it is about reigniting an individual's innate capacity for self-determination and empowering them to reclaim control over their health narrative.

4. The social fabric: connectivity and belonging

No individual exists in isolation. Our well-being is intricately woven into the fabric of our relationships and our sense of belonging. Connectivity, in this framework, refers both to the internal integration of our physiological and neural systems and, crucially, to our social bonds and sense of community. Humans are fundamentally social beings, and our brains are wired for connection.

From a neurobiological perspective, social connection engages a sophisticated "social brain network," including the temporoparietal junction (TPJ), superior temporal sulcus (STS), and medial prefrontal cortex (mPFC), which are vital for understanding others' intentions and fostering empathy [14]. The release of oxytocin, often dubbed the "love hormone," plays a critical role in bonding, trust, and social recognition, further cementing our desire for connection [15]. Moreover, the vagus nerve, a key component of the parasympathetic nervous system, is deeply involved in regulating social engagement and emotional

Table 1
The three pillars of lifestyle medicine.

Pillar	Definition	Neurobiological Mechanism	Clinical Benefits	Practical Tools
Presence	Mindful, non-judgmental awareness of internal bodily states (interoception) and external environment.	Anterior Insula, ACC, Brainstem: Integration of visceral, somatic, and affective signals; Interoceptive Predictive Coding.	Enhanced self-regulation, emotional intelligence, reduced allostatic load, informed decision-making.	Mindfulness Meditation, Body Scan, Mindful Eating, Breathwork, Yoga.
Agency	Belief in one's capacity to act intentionally and effectively, translating insight into behavior.	Prefrontal Cortex (dlPFC, vmPFC), Basal Ganglia, Default Mode Network (DMN): Goal setting, planning, self-efficacy, active inference.	Increased motivation, sustained behavior change, improved self-efficacy, greater internal locus of control.	Motivational Interviewing, SMART Goals, Self-Monitoring, Value Clarification, Cognitive Restructuring.
Connectivity	Integration of internal physiological systems, neural networks, and social belonging with others.	Mirror Neuron System, Oxytocin System, Social Brain Network (TPJ, STS, mPFC), Vagus Nerve: Empathy, bonding, social buffering.	Reduced social isolation, enhanced empathy, stronger social support, improved stress resilience, sense of belonging.	Group Therapy, Community Engagement, Volunteering, Empathy Training, Shared Activities.

responses, highlighting the physiological underpinnings of our relationships [16]

The experience of social connection acts as a powerful buffer against stress and promotes resilience. When we feel seen, understood, and supported by others, our body's physiological response to stress is attenuated—our cortisol levels decrease, and our immune function improves [17]. Conversely, social isolation and loneliness are not merely unpleasant states; they are potent risk factors for chronic disease, comparable in impact to smoking or obesity [18]. A person experiencing profound loneliness might struggle with unhealthy eating patterns, not out of hunger, but as a misguided attempt to fill an emotional void.

Integrating connectivity into LM involves recognizing that health behaviors are often influenced by social norms, support systems, and community resources. It's about understanding that a patient's adherence to a dietary plan might be strengthened or undermined by their family's eating habits, or that their commitment to exercise is sustained by a walking group. By actively fostering a sense of belonging and community, LM can leverage the inherent human drive for connection to create a supportive environment where healthy choices are the norm, not the exception. This means building interventions that aren't just individualistic but are deeply embedded within social contexts, recognizing that our health is a collective endeavor.

5. Clinical relevance: when the internal compass falters

When Presence, Agency, and Connectivity are compromised, the consequences ripple across both mental and physical health. The individual loses their internal compass, struggling to interpret bodily signals, act effectively, or find solace in social bonds. Let's explore how these deficits manifest in common clinical conditions. This framework is transdiagnostic, applying across the DSM-5 and ICD-11 spectrums, though implementation varies by clinical severity and comorbidities.

5.1. Clinical scenarios: applying the framework across the Diagnostic spectrum

Scenario A: Anxiety Disorders (The Alarm State)

Clinical Presentation: A patient with Generalized Anxiety Disorder (GAD) interprets benign physiological signals (tachycardia) as catastrophic.

Framework Application: LM interventions focus on interoceptive reappraisal through grounding techniques. By reducing the "interoceptive prediction error," the patient builds agency to initiate coping strategies instead of avoidance. This applies broadly across anxiety disorders in DSM-5/ICD-11, though in severe cases like postpartum psychosis, psychiatric intervention takes precedence over LM alone.

Scenario B: Depression **(The Emotional Numbness)**

Clinical Presentation: A patient with Major Depressive Disorder reports anhedonia and apathy.

Framework Application: Behavioral activation reconnects sensory experiences, fostering agency through goal setting. Effective alongside antidepressants, but LM complements rather than replaces treatment for comorbid conditions.

Scenario C: Eating Disorders (The Control Paradox)

Clinical Presentation: An individual with Anorexia Nervosa suppresses hunger cues.

Framework Application: Mindful eating rebuilds trust in internal signals, reclaiming agency. Applicable to DSM-5 spectrums, but integrated with therapy for severe malnutrition.

Scenario D: Chronic Pain (The Fear-Avoidance Cycle)

Clinical Presentation: A patient with fibromyalgia avoids movement due to hyper-vigilance.

Table 2
Disorders associated with deficits of presence and agency.

Clinical Condition	Presence Deficit	Agency Deficit	Underlying Mechanism	LM Intervention Indicated
Anxiety Disorders	Attentional bias to threat, misinterpretation of benign internal sensations (e.g., heart palpitation as heart attack), reduced interoceptive accuracy.	Avoidance behaviors, catastrophic thinking, difficulty initiating coping strategies, feeling overwhelmed and powerless.	Hyperactive amygdala, dysregulated insula-ACC circuits, impaired top-down control from PFC, maladaptive predictive coding where body signals are habitually predicted as threatening. Chronic sympathetic overdrive.	Mindfulness-Based Stress Reduction (MBSR), Body Scan, Exposure Therapy (with interoceptive focus), Cognitive Restructuring, CBT for Anxiety.
Depression	Numbness, emotional anhedonia, reduced awareness of positive bodily sensations, poor energy perception, somatic complaints.	Apathy, low motivation, difficulty initiating self-care, learned helplessness, rumination leading to inaction.	Hypoactivity in reward circuits (VTA, NAcc), reduced functional connectivity between DMN and task-positive networks, impaired prefrontal regulation of emotion, allostatic load from chronic stress leading to depletion of executive resources.	Behavioral Activation, Goal Setting, Exercise Prescription, Value Clarification, Mindfulness-Based Cognitive Therapy (MBCT), Social Engagement, Light Therapy.
Eating Disorders	Disconnection from hunger/satiety cues, distorted body image, reduced interoceptive accuracy (especially hunger/fullness).	Restrictive behaviors, compulsive eating, purging, feeling "out of control" around food, inability to trust internal signals for nourishment.	Dysregulation in hypothalamus-pituitary-adrenal (HPA) axis, impaired reward processing related to food, altered gut-brain axis communication, dysfunctional insula activity in processing taste/satiety, maladaptive learning associating food with guilt/control.	Mindful Eating, Intuitive Eating, Nutrition Education (non-judgmental), Exposure to Feared Foods, Body Image Work, Self-Compassion, Group Support.
Chronic Pain	Hyper-vigilance to pain signals, catastrophizing of sensations, reduced ability to differentiate pain from normal bodily feelings.	Passivity, avoidance of movement, reduced engagement in valued activities, feeling trapped by pain, reliance on external pain management.	Central sensitization, neuroplastic changes in somatosensory cortex and insula, altered descending pain modulation, fear-avoidance model, impaired prefrontal control over pain perception, heightened salience network activity for pain.	Graded Activity Exposure, Acceptance and Commitment Therapy (ACT), Pain Neuroscience Education, Mindfulness for Pain Management, Gentle Movement/Yoga, Pacing Strategies.
Metabolic Disorders	Reduced awareness of satiety/hunger, impaired perception of energy levels, poor detection of stress-induced physiological changes.	Difficulty sustaining dietary changes or exercise regimens, reliance on external rules for eating/activity, feeling overwhelmed by lifestyle modification demands.	Dysregulation of metabolic hormones (insulin, leptin, ghrelin), chronic inflammation, altered gut microbiome, impaired reward system responses to healthy foods, stress-induced eating behaviors, social determinants of health limiting access to healthy choices.	Mindful Eating, Personalized Nutrition Coaching, Regular Exercise (structured and unstructured), Stress Management Techniques, Sleep Hygiene, Peer Support Groups.

Table 3
Practical strategies organized by context.

Context	Objective	Technique	Duration	Frequency	Expected Results
Mind-Body	Enhance Presence (Interoception, Self-Awareness)	Mindfulness-Based Stress Reduction (MBSR) Program, Body Scan Meditation, Yoga/Tai Chi, Guided Breathwork.	8-week program; 10-45 min sessions	Daily practice; weekly classes	Improved interoceptive accuracy/awareness, reduced stress, enhanced emotional regulation, greater self-compassion.
Nutrition	Foster Presence (Mindful Eating) & Agency (Healthy Food Choices)	Mindful Eating Exercises (e.g., raisin meditation, sensory awareness during meals), Intuitive Eating principles.	5-20 min per meal	Daily, with each meal	Better recognition of hunger/satiety cues, reduced overeating, improved digestion, more satisfying relationship with food.
Physical Activity	Build Agency (Self-Efficacy, Motivation) & Presence (Bodily Awareness)	SMART Goal Setting for Exercise, Progressive Resistance Training, Nature Walks, Mindful Movement practices.	30-60 min	3-5 times/week	Increased physical strength/endurance, improved mood, enhanced body awareness, sustained exercise adherence.
Stress Management	Cultivate Presence (Emotional Regulation) & Agency (Coping Skills)	Journaling, Progressive Muscle Relaxation (PMR), Cognitive Behavioral Therapy (CBT) Techniques, Time Management.	10-30 min	Daily/As needed	Reduced perceived stress, improved sleep quality, effective coping strategies, greater emotional resilience.
Social Connection	Strengthen Connectivity (Belonging, Support)	Group Therapy/Support Groups, Volunteering, Community Activities, Establishing Social Routines.	60-90 min	Weekly/Monthly	Decreased loneliness, enhanced social support, improved mental health, sense of purpose, shared accountability.
Behavior Change	Empower Agency (Motivation, Self-Efficacy)	Motivational Interviewing (MI), Value Clarification Exercises, Habit Stacking, Relapse Prevention Strategies.	15-60 min	Ongoing	Increased internal motivation, sustained behavioral change, stronger belief in self, effective management of setbacks.
Assessment	Measure Presence (Interoception) & Agency (Self-Efficacy)	Multidimensional Assessment of Interoceptive Awareness (MAIA), General Self-Efficacy Scale (GSES).	10-20 min	Pre/Post Intervention	Objective tracking of progress, identification of specific deficits, personalization of interventions.

Framework Application: Pain neuroscience education differentiates pain from normal sensations, restoring agency. Effective alongside multimodal pain management for comorbidities.

Scenario E: Metabolic Disorders (The Metabolic Disconnect)

Clinical Presentation: A patient with Type 2 Diabetes struggles with satiety perception.

Framework Application: Mindful eating and stress management empower conscious choices. Applicable across ICD-11, but addresses social determinants limiting access to healthy options.

6. Cultivating resilience: practical strategies in lifestyle medicine

The good news is that Presence, Agency, and Connectivity are not fixed traits; they are dynamic capacities that can be intentionally cultivated. LM offers a rich toolkit of strategies that directly address these pillars, guiding individuals toward greater self-awareness, empowerment, and belonging.

Mind-Body Practices for Presence: Mindfulness-Based Stress Reduction (MBSR) and practices like body scans or mindful breathing are potent tools for enhancing interoceptive awareness [19]. By directing attention inward, individuals learn to observe bodily sensations without judgment, building a richer, more accurate map of their internal landscape. Yoga and Tai Chi further integrate mindful movement with breath, fostering a deeper connection between mind and body.

Motivational Interviewing for Agency: Motivational Interviewing (MI) is a collaborative, person-centered counseling style that elicits and strengthens an individual's motivation for change [20]. Instead of dictating solutions, MI guides individuals to articulate their own reasons for change, enhancing their sense of autonomy and fostering an internal locus of control—the very essence of agency. Setting SMART (Specific, Measurable, Achievable, Relevant, Time-bound) goals also provides a clear path for individuals to experience incremental successes, reinforcing their self-efficacy.

Fostering Connectivity: LM interventions often incorporate group-based activities, such as shared cooking classes, walking groups, or support forums. These settings naturally facilitate social interaction and build a sense of community, harnessing the power of collective support to sustain individual behavioral changes. Encouraging volunteering or participation in local community initiatives also extends the sense of belonging beyond formal health settings.

Assessment Tools: Tools like the Multidimensional Assessment of Interoceptive Awareness (MAIA) scale [21] can help clinicians identify specific interoceptive deficits, guiding personalized interventions. Similarly, self-efficacy scales can track a patient's belief in their ability to enact change, allowing for targeted strategies to build confidence.

7. Discussion

The Symphony of Self-Regulation The framework of Presence, Agency, and Connectivity offers a robust, neuropsychobiologically informed approach to Lifestyle Medicine. It moves beyond a reductionist view of lifestyle interventions as mere prescriptions, instead seeing them as profound opportunities for self-discovery and empowerment. When we truly understand the neural mechanisms linking internal awareness (Presence), intentional action (Agency), and social bonding (Connectivity), we unlock a more holistic and sustainable path to health.

The strength of this framework lies in its synergistic nature. Enhanced Presence provides richer, more accurate data from the body, enabling more informed decision-making. This, in turn, fuels Agency, as individuals feel more capable and confident in acting upon their internal wisdom. Stronger Agency then reinforces positive health behaviors, which are often sustained and amplified within supportive social networks and communities (Connectivity). It's a virtuous cycle, where each pillar strengthens the others, leading to a profound and lasting transformation.

However, challenges remain. Research needs to further delineate personalized interventions for individuals with specific neuropsychobiological deficits. How can we best tailor interoceptive training for someone with a dysregulated insula versus someone with a compromised prefrontal cortex? Furthermore, while the neurobiological underpinnings are increasingly understood, translating this complex science into accessible, scalable, and equitable interventions for diverse populations is crucial. Integrating digital health tools, biofeedback, and neurofeedback approaches within this framework holds significant promise for the future. We must also acknowledge the profound impact of social determinants of health, which can significantly impede an individual's capacity for Presence, Agency, and Connectivity, regardless of their intrinsic capabilities. A truly humanized approach to LM must address these broader systemic inequalities.

Furthermore, the synergy between Presence and Agency suggests a neurobiological feedback loop that is essential for habit formation. When an individual enhances their interoceptive accuracy (Presence), they become more sensitive to the immediate physiological rewards of healthy behaviors—such as the post-exercise endorphin release or the improved glycemic stability following a balanced meal. This heightened sensory feedback provides a more salient "reward signal" to dopaminergic pathways, which in turn strengthens perceived self-efficacy (Agency). By making the internal rewards of Lifestyle Medicine more "visible" to the brain, we move beyond reliance on delayed gratification, anchoring behavioral change in the immediate, felt experience of well-being.

his framework also aligns with and expands upon Self-Determination Theory, which posits that autonomy, competence, and relatedness are fundamental psychological needs [22,23]. Our proposed pillars of Agency, Presence, and Connectivity map directly onto these constructs. Agency provides a neurobiological substrate for autonomy; Presence facilitates competence in navigating one's internal states; and Connectivity fulfills the need for relatedness. By integrating these needs with their neuropsychobiological underpinnings, Lifestyle Medicine can transition from a prescriptive model—where the provider dictates behavior—to a concordance-oriented approach emphasizing shared goals and patient partnership [24]. In this shift, the patient is not merely compliant but becomes an active collaborator whose internal locus of control drives the therapeutic process.

From a clinical implementation perspective, the role of the healthcare provider must evolve into that of a facilitator of agency. This requires a move away from purely didactic education toward strategies that prioritize interoceptive inquiry and motivational alignment. For instance, instead of advising a patient to "reduce stress," a Presence-informed approach would teach the patient to recognize early somatic markers of stress (e.g., muscle tension or shallow breathing). Once these signals are identified, Agency-building strategies, such as habit stacking or cognitive reappraisal, can be deployed to intervene before the stress response escalates. This granular approach ensures that interventions are not only evidence-based but also personally meaningful and embodied in lived experience.

Finally, the ethical and equity implications of this framework deserve explicit attention. While Presence and Agency are human capacities, they are strongly shaped by allostatic load—i.e., the cumulative physiological burden of chronic stress [25,26]—which is disproportionately driven by adverse social determinants of health [27,28]. Individuals living in high-stress, low-resource environments may develop interoceptive numbing or diminished agency as adaptive responses to systemic instability. Therefore, Lifestyle Medicine interventions must be culturally responsive, context-sensitive, and realistic about structural barriers. Strengthening Connectivity at community and systems levels is not simply a social goal but a biological necessity to buffer chronic stress and create the conditions in which individuals can re-engage with their internal compass and reclaim their health trajectories.

8. Conclusion

The Human Heart of Lifestyle Medicine Lifestyle Medicine, at its core, is about empowering individuals to live healthier, more vibrant lives. By focusing on Presence, Agency, and Connectivity, we tap into fundamental human capacities that are essential for self-regulation and sustainable health transformation. It's about moving from passively receiving health advice to actively orchestrating one's well-being, guided by an attuned internal compass, empowered by intentional action, and supported by meaningful connections. This neuropsychobiological framework offers a richer, more human-centric understanding of how individuals can not just manage disease, but truly flourish, creating a healthier future one conscious breath, one intentional choice, and one heartfelt connection at a time. It reminds us that at the heart of science, there is always a human story waiting to be heard and transformed.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

none

References

[1] Lianov LS, Kennedy T. American college of lifestyle medicine: history and future. Am J Lifestyle Med 2018;12(5):361–8.
[2] Craig AD. How do you feel? Interoception: the sense of the physiological condition of the body. Nat Rev Neurosci 2002;3(8):655–66.
[3] Seth AK. Interoceptive inference, emotion, and the embodied self. Trends Cognit Sci 2013;17(11):565–73.
[4] Barrett LF, Simmons WK. Interoceptive predictions in the brain. Nat Rev Neurosci 2015;16(7):419–29.
[5] Critchley HD, Harrison NA. Insular cortex and the regulation of bodily homeostasis. Curr Opin Neurobiol 2013;23(3):384–92.
[6] Murphy J, Catmur C, Bird G. Atypical interoception and its relationship to alexithymia in autism. J Neurosci 2017;37(20):5407–14.
[7] Friston K. The free-energy principle: a unified brain theory? Nat Rev Neurosci 2010;11(2):127–38.
[8] Rushworth MFS, Noonan MAP, Boorman ED, Walton ME, Behrens TE. Frontal cortex and reward-guided learning and decision-making. Neuron 2011;70(6):1054–69.
[9] Bandura A. Self-efficacy: toward a unifying theory of behavioral change. Psychol Rev 1977;84(2):191–215.
[10] Schultz W. Dopamine reward prediction-error signalling: a two-component response. Nat Rev Neurosci 2016;17(3):183–95.
[11] Wise RA, Robble MA. Dopamine and addiction. Annu Rev Psychol 2020;71:79–106.
[12] Volkow ND, Koob GF, McLellan AT. Neurobiologic advances from the brain disease model of addiction. N Engl J Med 2022;388(4):156–62.
[13] Maier SF, Seligman ME. Learned helplessness: theory and evidence. J Exp Psychol Gen 1976;105(1):3–46.
[14] Lieberman MD. Social: why our brains are wired to connect. Crown; 2013.
[15] Young WS. Oxytocin and social bonding. Trends Cognit Sci 2009;13(11):473–8.
[16] Porges SW. The polyvagal perspective. Biol Psychol 2007;74(2):116–43.
[17] Uchino BN. Social support and physical health: understanding the health consequences of relationships. Yale University Press; 2004.
[18] Holt-Lunstad J, Smith TB, Baker M, Harris T, Stephenson D. Loneliness and social isolation as risk factors for mortality: a meta-analytic review. Perspect Psychol Sci 2015;10(2):227–37.
[19] Kabat-Zinn J. An outpatient program in behavioral medicine for chronic pain patients based on the practice of mindfulness meditation: theoretical considerations and preliminary results. Gen Hosp Psychiatry 1982;4(1):33–47. https://doi.org/10.1016/0163-8343(82)90026-3.
[20] Miller WR, Rollnick S. Motivational interviewing: helping people change. third ed. Guilford Press; 2012.
[21] Mehling WE, Price C, Daubenmier JJ, Acree M, Bartmess E, Stewart A. The multidimensional assessment of interoceptive awareness (MAIA). PLoS One 2012;7(11):e48230.
[22] Deci EL, Ryan RM. The "what" and "why" of goal pursuits: human needs and the self-determination of behavior. Psychol Inq 2000;11(4):227–68.
[23] Ryan RM, Deci EL. Self-determination theory: basic psychological needs in motivation, development, and wellness. Guilford Press; 2017.

[24] Royal Pharmaceutical Society of Great Britain. From compliance to concordance: achieving shared goals in medicine taking. Royal Pharmaceutical Society of Great Britain; 1997.
[25] McEwen BS, Stellar E. Stress and the individual: mechanisms leading to disease. Arch Intern Med 1993;153(18):2093–101.
[26] McEwen BS. Protective and damaging effects of stress mediators. N Engl J Med 1998;338(3):171–9.
[27] World Health Organization. Closing the gap in a generation: health equity through action on the social determinants of health. World Health Organization; 2008.
[28] Marmot M, Friel S, Bell R, Houweling TAJ, Taylor S, Commission on Social Determinants of Health. Closing the gap in a generation. Lancet 2008;372(9650):1661–9.